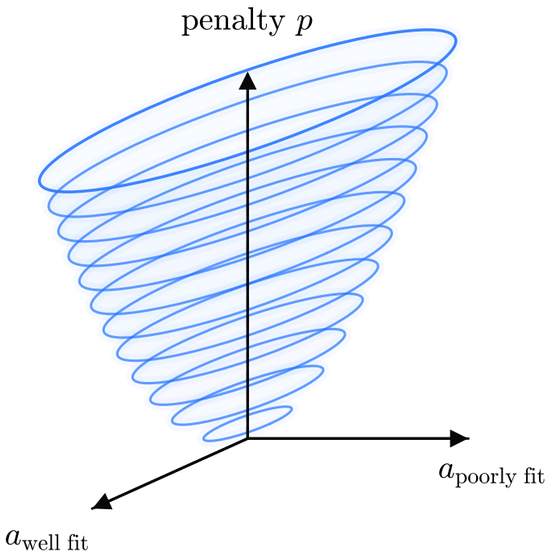
Continual Learning as Optimization Constraint via Moreau-Yosida Regularization

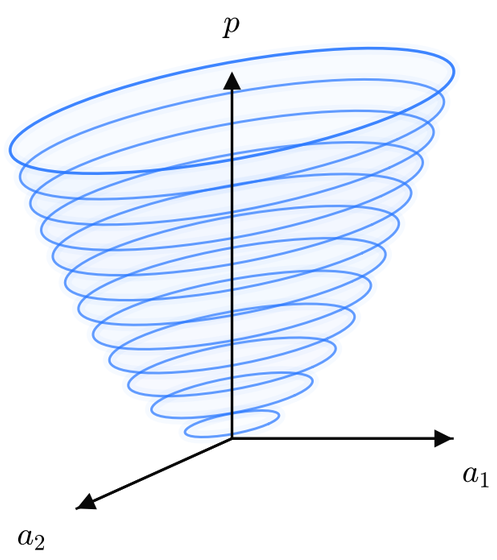
EF-EWC



$$p = \lambda(w_p(a_{\text{poorly fit}})^2 + w_w(a_{\text{well fit}})^2), \quad w_p \gg w_w$$

EF-EWC favors high-gradient samples

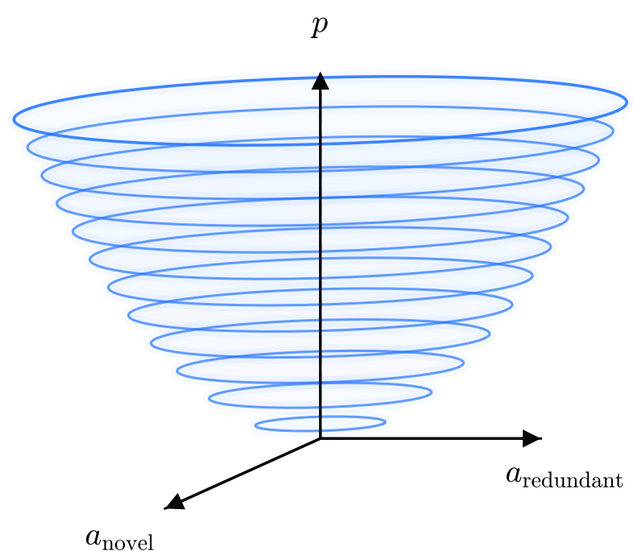
Uniform IEWC



$$p = \lambda((a_1)^2 + (a_2)^2)$$

Uniform IEWC treats all samples equally

GSS-IEWC



$$p = \lambda(h_r(a_{\text{redundant}})^2 + h_n(a_{\text{novel}})^2), \quad h_r \ll h_n$$

GSS-IEWC favors novel / diverse samples